

Project 3: Supply chain resiliency with RAG and KG

Concept

Develop a knowledge graph that maps a **medical manufacturing supply chain**, from raw material origins and third-party suppliers to manufacturing plants, distribution centres, and final product destinations.

Integrate this KG with a RAG component that can ingest real-time news feeds, geopolitical updates, weather alerts, and supplier performance data.

The system should answer complex, multi-hop questions about supply chain vulnerabilities, predict potential disruptions, and assess their impact on product availability and delivery.

This project offers a highly practical application of RAG and KGs, directly addressing a critical business need for any medical manufacturing company: maintaining an uninterrupted supply of vital products. The ability to perform multi-hop queries and discover hidden patterns within a complex global network makes for a powerful demonstration.

Healthcare Value:

- **Ensured Product Availability:** Minimize stockouts of critical medical devices and pharmaceuticals.
- **Proactive Risk Mitigation:** Identify and address potential supply chain disruptions before they escalate.
- **Optimized Inventory Management:** Make informed decisions about buffer stock and alternative sourcing.
- **Regulatory Compliance:** Track the origin of materials for compliance with regulations (e.g., country of origin labelling, ethical sourcing).

Key Features for Demo:

- **Impact Analysis of Disruptions (Direct Question):** "Vietnam is undergoing a power outage running into multiple days. Which products will be delayed in shipping?"
 - The system would trace dependencies: Vietnam (region) → specific suppliers in Vietnam → raw materials/components produced by those suppliers → medical products that use those materials → manufacturing plants that produce those products → distribution centers → final shipping routes.

- The RAG component would be crucial here to ingest and interpret the news about the power outage, understand its geographical scope and duration, and then link it to the KG.
- **Supplier Risk Assessment (Nested Query):** "Identify all critical raw materials sourced from Supplier X, which also has a major manufacturing plant located in an earthquake-prone region. What are the alternative suppliers for these materials, and what is their lead time?"
 - This requires querying supplier locations, material types, geographical risk profiles, and then alternative supplier data.
- **Geopolitical Impact Analysis (Nested Query):** "If there's a new trade tariff imposed on goods from Country Y, which of our high-volume diagnostic kits will see a significant increase in production cost, and what are the alternative sourcing options for their key components outside of Country Y?"
 - This involves country-specific trade policies, product bill of materials, and alternative supplier mapping.
- **Discovery of Not-So-Easy-to-Find Patterns:**
 - **Cascading Failures:** Uncover subtle chains of dependencies that could lead to cascading failures across the supply chain, even if individual components seem robust. For example, "A specific type of plastic resin, sourced from a single supplier in Region A, is used in three different critical components across five different medical devices. What is the cumulative impact if that supplier faces a 2-week production halt due to civil unrest in Region A?"
 - **Hidden Bottlenecks:** Identify obscure single points of failure in the supply chain that are not immediately obvious from top-level diagrams, such as a specialized piece of equipment only available from one manufacturer globally, affecting multiple product lines.
 - **Correlation of seemingly unrelated events:** Discover correlations between seemingly unrelated global events (e.g., a specific weather pattern in one region consistently leading to delays from a particular logistics provider, impacting certain product categories).
 - **Ethical Sourcing Anomalies:** Identify instances where a raw material's country of origin, based on supplier declarations, doesn't align with broader geopolitical or labour practice reports, suggesting a potential compliance risk.

Suggested approach

"A power outage running into multiple days in **Vietnam**. Which **medical products** will be **delayed in shipping**?"

- "power outage," "delayed in shipping" (semantic search on news feeds, operational reports to interpret the disruption).
- "Vietnam," "medical products" (keyword search for specific locations, or direct KG query for product categories).
- KG provides the explicit chain: country → supplier → raw material → component → product → shipping route.

Hybrid: RAG ingests and interprets the news event, then uses keyword/semantic identification of the affected region/event. This information is then used to trigger a multi-hop traversal in the KG to identify all dependent suppliers, materials, products, and shipping delays.

Data Sources

- **Synthetic Data Generation**
 - **Supply Chain Graph Data:** Generate realistic data for:
 - Raw materials (type, quantity, origin, quality parameters).
 - Suppliers (location, certifications, capacity, lead times, historical performance, tiering).
 - Manufacturing plants (location, production capacity, products manufactured).
 - Distribution centers (location, storage capacity, shipping routes).
 - Products (Bill of Materials - BOM, production volume, shipping destinations).
 - Geopolitical risks (synthetic risk scores for regions, types of events like natural disasters, political instability).
 - **Simulated Real-time Data:** Simulate news feeds (e.g., power outages, port strikes, natural disasters in specific regions), geopolitical updates, and supplier performance metrics.
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- **Datasets for RAG context and risk assessment:**

- **Global News APIs:** For real-time event monitoring (e.g., newsapi.org, newsdata.io, gnews.io, Serper/Google News API).
- **Simulate news feeds** (e.g., power outages, port strikes, natural disasters in specific regions), geopolitical updates, and supplier performance metrics.
- **Internal Data (Simulated):**
 - Purchase orders, invoices, shipping manifests.
 - Inventory levels, production schedules.
 - Supplier audit reports.
 - Product specifications and bills of materials.

Data generation tips

1. Use Claude to generate python script that will create the synthetic datasets needed for above (you can copy paste the entire text above and ask it to generate the needed data generation script with the patterns in there needed to answer the questions mentioned above)
2. Use Claude to generate the neo4j DDL script that will contain the cypher queries to create the graph nodes and edges
3. Following could be from external APIs/loaded into memory from JSON flat file
 - News feeds** - For real-time ingestion via RAG
 - Purchase orders** - Transactional data
 - Inventory levels** - Frequently changing data

A reference dataset is provided in the project3-data folder

Dataset Summary:

generated_at: 2025-05-29T14:10:35.403976

total_suppliers: 150

total_materials: 75

total_plants: 75

total_distribution_centers: 50

total_products: 36

total_countries: 24

data_version: 1.0

Total files created: 15

Key Features in the dataset:

Graph Structure:

- **Multi-hop queries** supported (Vietnam → Suppliers → Materials → Products)
- **Relationship properties** (quantities, costs, lead times)
- **Geographic hierarchy** (Country → Region)
- **Risk scoring** integrated into country nodes

Sample Queries Included:

1. **Vietnam Impact Analysis** - Find all products affected by Vietnam disruptions
2. **Earthquake Risk Assessment** - Suppliers in high-risk zones
3. **Single Points of Failure** - Materials with only one supplier
4. **Alternative Sourcing** - Find backup suppliers for critical materials